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|----------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |          | $F \propto \sqrt{E}$                                                                                                                                                                                   |
| <b>4</b> | <b>C</b> | <p>Elastic collision <math>\rightarrow</math> by relative speed of approach = relative speed of separation</p> <p>Taking rightward as positive,</p> $u_1 - (-u_2) = v_2 - v_1$ $u_1 + u_2 = v_2 - v_1$ |
| <b>E</b> | <b>D</b> | Method 1 (by calculation):                                                                                                                                                                             |